

MERGE 4 DESIGN LLC × L.E.E.P., INC.

FORWARD HOME BUILDING SYSTEM

Project Specification — Four-Tier Performance Framework + Factory Assembly Program

Version 0.2 — ICF Foundation Update · June 2026 · Supersedes May 2026 foundation specification

**Change summary:** Superior Walls Xi Plus (TER 2211-02) reclassified to suburban/rural sites only — crane + large staging area required, incompatible with dense urban infill. Fox Blocks 6" ICF stem wall (ER-6163) adopted as T2–T4 urban infill primary foundation specification. Fox Blocks 8" ICF adopted for T4 maximum performance. LiteForm FPSF LiteFoot (IRC R403.3) added as T1 budget path and tight-site alternative on all tiers. ICF foundation eliminates separate WRB, damp-proofing, and thermal bridge at sill. L.E.E.P. crew trainable — no licensed masonry trade required. Concrete pump subcontract required for pour.

**Supplier contact action:** Contact Fox Blocks (800-369-9162 / foxblocks.com) and NUDURA (866-468-6832 / nudura.com) for contractor pricing on 6" and 8" forms. Contact LiteForm (402-345-4380 / liteform.com) for FPSF LiteFoot pricing. Add to Supplier Cost Tracker SUPPLIER\_DIR tab. Get concrete pump subcontractor pricing per pour (target: single half-day event per unit).

2B. Wall Assembly Matrix — Updated

Foundation specification: ICF is the primary specification for urban infill sites on all tiers where crane access and large staging areas are not feasible. Superior Walls Xi Plus remains available for suburban and rural site conditions where staging is not constrained. LiteForm FPSF is the T1 budget path and the preferred alternative on any site requiring minimum excavation.

| Element                                         | T1 — Code+                                                                           | T2 — Zero                                                                                                     | T3 — Positive                                             | T4 — Harvest                                       |
|-------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------|
| ZONE 1 — FOUNDATION (inside → outside)          |                                                                                      |                                                                                                               |                                                           |                                                    |
| Sill anchor / base track                        | Simpson MASOZ · ESR-2555 · CFS base track AISI S240                                  | Same + SIGA TEAKstrip sill capillary break                                                                    | Same as T2                                                | Same as T2                                         |
| Foundation wall — urban infill (primary)        | CMU / poured 3,500 psi · R-10 EPS under slab · R-10 edge · 15-mil VR                 | Fox Blocks 6" ICF stem wall · ER-6163 · R-22 continuous · no separate WRB/damp-proof · concrete pump for pour | Same as T2 — Fox Blocks 6" ICF                            | Fox Blocks 8" ICF · R-25+ · 4-hr ASTM E119         |
| Foundation wall — suburban/rural sites only     | —                                                                                    | Superior Walls Xi Plus · TER 2211-02 · requires crane + large staging area                                    | Same as T2 (suburban/rural only)                          | Same as T2 (suburban/rural only)                   |
| Foundation — budget path / all tight sites      | LiteForm FPSF LiteFoot · IRC R403.3 · stay-in-place EPS · less excavation + concrete | Same — FPSF viable on all tiers as tight-site alt.                                                            | Same                                                      | Same                                               |
| Separate WRB / damp-proof (ICF T2–T4)           | Required — separate trade ~\$2.40/SF                                                 | Not required — EPS face serves as damp-proof                                                                  | Not required                                              | Not required                                       |
| ZONE 2 — WALL PANEL ASSEMBLY (inside → outside) |                                                                                      |                                                                                                               |                                                           |                                                    |
| Interior finish                                 | ½" Type X GWB — factory hung at hub · tape+mud on site                               | Same · fluid-applied at all electrical penetrations                                                           | Same + Pro Clima Intello+ airtight membrane at junctions  | Same as T3                                         |
| CFS spine (point loads + hold-downs)            | 600S162-54 · AISI S240 · ESR-4205                                                    | Same                                                                                                          | Same                                                      | Same                                               |
| Structural wall panel                           | ACME 6.5" SIP — Neopor GPS / OSB · R-23 · ESR-4689                                   | Same SIP · Rockwool Comfortbatt R-23 mineral wool cavity upgrade                                              | Same SIP                                                  | Same SIP                                           |
| WRB + CI layer                                  | Barricade Thermo-Brace SIB R-5 · TER 1507-09 · WRB integrated · GREENGUARD Gold      | Same + Prosoco R-Guard at all penetrations                                                                    | Same + continuous air barrier lapped at foundation + roof | Same — critical for island-mode battery efficiency |

| Element                                                                     | T1 — Code+                                                                         | T2 — Zero                                                              | T3 — Positive                                                                                  | T4 — Harvest                                       |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Outboard CI (T3–T4 only)                                                    | —                                                                                  | —                                                                      | DuPont ArmorWall Plus FR SIS 2¼" R-15 · MgO/PUR · ESL-1302 · NFPA 285 · direct cladding attach | ArmorWall Plus FR SIS 3¼" R-21 · ESL-1302 · max CI |
| Party wall / fire separation                                                | 8" CMU continuous foundation → underside SIP roof · STC ≥ 50                       | Same + Thermafiber 4" + Hilti FS-ONE MAX · STC ≥ 55                    | Same + Huber EXACOR MgO face · ESR-4635 · 2-hr E119 · STC ≥ 60                                 | Same as T3                                         |
| CLT-to-CFS bearing connector                                                | Rothoblaas ALUMEGA · ESR-4546 / AC233<br>Rothoblaas HBS PLATE · ETA-11/0030        | Same                                                                   | Same — C4: HBS PLATE EVO / AC257                                                               | Same — A4 stainless: ALUMEGA + HBS PLATE A4        |
| Hold-downs                                                                  | Simpson HTT4 · ESR-1188                                                            | Simpson HTT5 · ESR-1188                                                | Simpson HTT5 · ESR-1188                                                                        | Simpson HTT22 · ESR-1188                           |
| SIP joint spline + airtight tape                                            | ACME spline per ESR-4689 · Rothoblaas SIGA Rissan at all SIP joints                | Same + SIGA FENTRIM IS at window/door perimeters                       | Same + SIGA MAJREX smart vapor retarder inboard                                                | Same as T3                                         |
| Stair cassette connector                                                    | Simpson SCLC · C-C-2024 · hub-built cassette lowered on site                       | Same                                                                   | Same                                                                                           | Same                                               |
| <b>ZONE 3 — EXTERIOR CLADDING (outside face)</b>                            |                                                                                    |                                                                        |                                                                                                |                                                    |
| Primary cladding                                                            | Allura® Lap 7" · factory primed · 50-yr warranty · 1" ventilated rainscreen on SIB | Allura® Panel Spectrum pre-finished · stainless clips · no field paint | Allura® Panel + Shakes · Spectrum · direct attach to ArmorWall MgO face — no hat channel       | Same as T3                                         |
| Trim + soffit                                                               | Allura Trim at all openings                                                        | Allura Soffit + Trim                                                   | Same as T2                                                                                     | Same as T2                                         |
| <b>ZONE 4 — SAPK CONNECTOR KIT SUMMARY</b>                                  |                                                                                    |                                                                        |                                                                                                |                                                    |
| Foundation zone                                                             | SST MASOZ G90 + HTT4                                                               | SST MASOZ + HTT5 + SIGA TEAKstrip                                      | Same as T2                                                                                     | SST MASOZ + HTT22                                  |
| Wall zone                                                                   | Rtb ALUMEGA + HBS PLATE · #14 SD AISI S240                                         | Same + Rtb SIGA Rissan tape                                            | Same + Rtb HBS PLATE EVO + SIGA FENTRIM IS                                                     | Same — all A4 stainless                            |
| Stair cassette                                                              | SST SCLC + SDS screws                                                              | Same                                                                   | Same                                                                                           | Same                                               |
| All items carry existing US listings                                        | No new UL or ICC testing required                                                  | AISI S240 / ESR-1188 / ESR-2555 / ESR-4205 / ESR-4546 / AC233          | Same + AC257 (EVO corrosion)                                                                   | Same — A4 SS coating                               |
| <b>ZONE 5 — EFFECTIVE WALL PERFORMANCE</b>                                  |                                                                                    |                                                                        |                                                                                                |                                                    |
| Effective R-value                                                           | R-21 to R-23                                                                       | R-22 to R-24                                                           | R-38+ (SIP + ArmorWall 2¾")                                                                    | R-44+ (SIP + ArmorWall 3¾")                        |
| Airtightness target                                                         | <2.0 ACH50 (auto — SIP)                                                            | <1.0 ACH50 — verified                                                  | <0.6 ACH50 — PHIUS verified                                                                    | <0.6 ACH50 — island-mode critical                  |
| <b>INSTALLED COST COMPARISON — \$/SF WALL (Mid-Atlantic, June 2026 ROM)</b> |                                                                                    |                                                                        |                                                                                                |                                                    |
| Traditional stick-frame / CMU basement                                      | \$38.50/SF                                                                         | \$52.80/SF                                                             | \$71.40/SF                                                                                     | \$84.60/SF                                         |
| Forward System — partial factory (incl. ICF foundation T2+)                 | \$36.20/SF                                                                         | \$44.60/SF                                                             | \$55.80/SF                                                                                     | \$62.40/SF                                         |
| ICF foundation savings vs CMU frost wall (T2+)                              | —                                                                                  | –\$4.00 to –\$6.00/SF                                                  | –\$4.00 to –\$6.00/SF                                                                          | –\$4.00 to –\$6.00/SF                              |
| MEP factory credit                                                          | –\$2.80/SF                                                                         | –\$5.40/SF                                                             | –\$7.20/SF                                                                                     | –\$9.80/SF                                         |
| <b>Installed cost/SF wall</b>                                               | <b>\$33.40/SF</b>                                                                  | <b>\$39.20/SF</b>                                                      | <b>\$48.60/SF</b>                                                                              | <b>\$52.60/SF</b>                                  |

| Element                                            | T1 — Code+      | T2 — Zero        | T3 — Positive    | T4 — Harvest     |
|----------------------------------------------------|-----------------|------------------|------------------|------------------|
| Savings vs stick-frame (with MEP + ICF foundation) | \$5.10/SF (13%) | \$17.60/SF (33%) | \$26.80/SF (38%) | \$36.00/SF (43%) |

**Foundation cost note:** ICF stem wall (T2–T4) eliminates the separate WRB/damp-proof trade (~\$2.40/SF), separate below-grade insulation (~\$3.40–\$4.80/SF), and the thermal bridge at the sill plate. Net installed cost is comparable to or below CMU frost wall when whole-wall trades are included. Concrete pump subcontract is a new line item (~\$400–\$800/unit at small residential scale) but is offset by the trades eliminated. ROM figures in the cost table above are wall assembly only; foundation-specific net savings are shown separately in the ICF savings row.